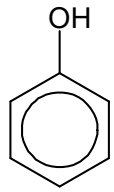
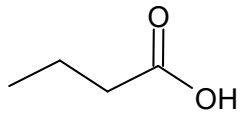
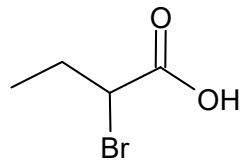
2 **(a)** Deduce the order of increasing acid strength of the following compounds. Explain your answer.



phenol



butanoic acid



2-bromobutanoic acid

[3]

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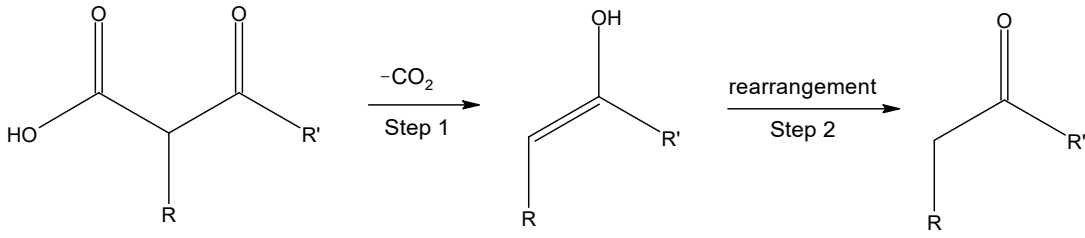
(b) Decarboxylation is the loss of carbon dioxide from a carboxylic acid group. It plays an important role in organic synthesis and occurs under specific thermal or catalytic conditions.

Two examples of decarboxylation are shown in Fig. 2.1.

Example 1:



Example 2:

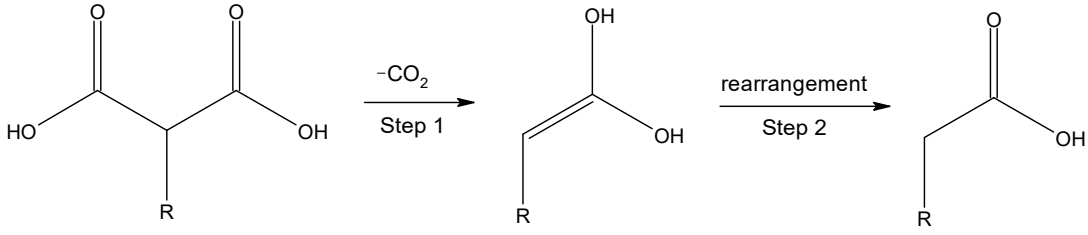
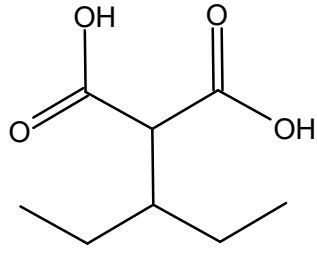


Fig. 2.1

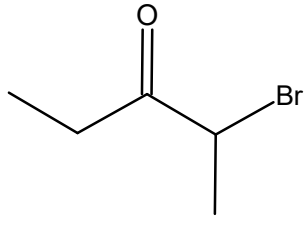
(i) Draw the structure of the organic product formed when compound **K** undergoes decarboxylation.



K

[1]

(ii) Two different acids, **M** and **N**, each can undergo decarboxylation to give **L**.



L

Suggest possible structures of **M** and **N**.

[2]

(iii) Compound **L** can also be synthesised from pentan-3-ol by the three-step route shown in Fig. 2.2

State the reagents and conditions required for each step and suggest structures for the organic compounds **Q** and **R**.

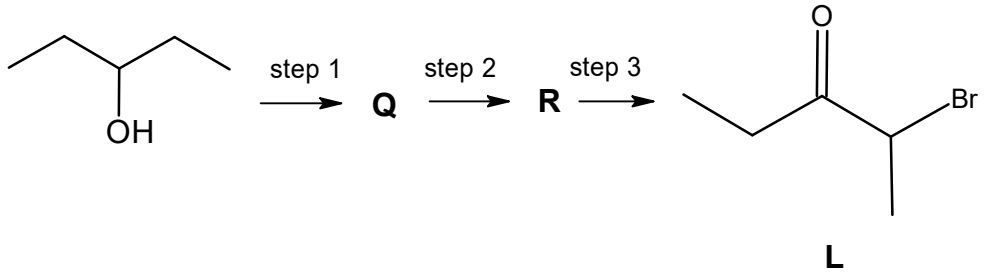


Fig. 2.2 **[5]**

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(c) The Kolbe electrolysis reaction involves the decarboxylative dimerisation of carboxylate ions to form alkanes at the anode by the following reaction.

$$2R–CO_2^- \rightarrow R–R + 2CO_2 + 2e^-$$

R = $–CH_3$, $–CH_2CH_3$, etc.

For example, the electrolysis of sodium ethanoate solution produces ethane at the anode.

(i) Suggest the products of the reaction at the cathode and construct an equation for the electrolysis of sodium ethanoate. **[2]**

(ii) Calculate the mass of ethane, C_2H_6, produced when a current of 2.0 A is passed through a solution of sodium ethanoate for 55 minutes. **[2]**

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(d) Group 2 ethanoates decompose when heated to form carbonates and propanone.

(i) Write an equation for the thermal decomposition of $Ba(CH_3COO)_2$. **[1]**

(ii) The trend in the thermal stability of Group 2 ethanoates is similar to that of Group 2 carbonates.

Describe and explain the trend in the thermal stability of Group 2 ethanoates. **[2]**

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